

AquaWatch

AI-Powered Smart Water Infrastructure Monitoring and Decision Support System

Domain Document

Final Year B.E. Project | Department of Computer Science and Engineering

1. Domain Overview

The proposed research is situated in the domain of Smart City and Municipal Water Infrastructure Management. Water infrastructure is a critical component of urban development, responsible for ensuring the continuous supply of clean water and the proper disposal of wastewater. Municipal corporations are responsible for maintaining water distribution pipelines, drainage systems, sewage networks, and other related infrastructure. Citizens report infrastructure failures such as pipeline leakages, waterlogging, sewage overflow, drainage blockages, and damaged water supply systems through municipal complaint portals, mobile applications, or helplines.

As urban populations continue to grow, the number of civic complaints has increased significantly, making manual complaint handling inefficient. Timely identification, classification, prioritization, and resolution of these complaints are essential to minimize water wastage, reduce environmental pollution, improve public health, and ensure efficient utilization of municipal resources.

2. Existing Domain Workflow

The current municipal complaint management process generally consists of the following stages:

- 1 A citizen identifies a water-related infrastructure issue.
- 2 The complaint is submitted through a web portal or mobile application along with an image and a textual description.
- 3 The complaint is stored in the municipal database.
- 4 Municipal officials manually verify and categorize the complaint.
- 5 The complaint is assigned to the appropriate maintenance department.
- 6 Maintenance personnel resolve the issue.
- 7 The complaint status is updated and communicated to the citizen.

Although some modern systems automate basic image classification, most of the decision-making process remains manual.

3. Challenges in the Domain

Current municipal water complaint management systems face several operational challenges:

- Large volumes of complaints increase the administrative workload.
- Duplicate complaints related to the same incident are processed independently.
- Complaint prioritization is performed manually without considering issue severity.
- Existing systems cannot automatically identify recurring infrastructure failure hotspots.
- Resource allocation is often inefficient due to the absence of intelligent decision support.
- Response time increases because authorities manually analyze complaint information.

These limitations reduce operational efficiency and delay the resolution of critical water infrastructure issues.

4. Role of Artificial Intelligence

Artificial Intelligence can significantly enhance municipal water complaint management by automating analysis and supporting decision-making. In the proposed AquaWatch system:

- **Computer Vision** automatically classifies water-related infrastructure issues from uploaded images.
- **Natural Language Processing** extracts meaningful information from complaint descriptions.
- **Multimodal Learning** combines image, text, GPS location, and timestamp information for comprehensive complaint understanding.
- **Similarity Search** identifies duplicate complaints.
- **Geospatial Clustering** detects complaint hotspots.
- **Severity Prediction** estimates the urgency of reported issues.
- **AI-based Prioritization** assists municipal authorities in allocating maintenance resources effectively.

5. Stakeholders

The primary stakeholders of the system are:

Citizens

- Report water-related civic issues.
- Track complaint status.
- Receive updates regarding complaint resolution.

Municipal Authorities

- Verify complaints.
- Prioritize maintenance activities.
- Monitor complaint analytics.
- Allocate resources efficiently.

Maintenance Teams

- Receive assigned complaints.
- Resolve reported issues.
- Update maintenance status.

City Administration

- Analyze infrastructure performance.
- Identify recurring problem areas.
- Support long-term urban planning using data analytics.

6. Expected Domain Impact

The proposed system aims to transform conventional complaint management into an intelligent decision-support platform by reducing manual effort, minimizing duplicate complaint processing, improving response time, optimizing resource allocation, and enabling data-driven municipal governance. Ultimately, AquaWatch contributes to the development of smart and sustainable cities by improving the efficiency of urban water infrastructure management.